

# Comprehensive Taxonomy of Theory of Equations & Vieta's Relations

A Master Categorization of 101 Problem Types with Representative Questions

Advanced Mathematics Pedagogy Series

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# 1 Introduction

The theory of equations and the algebraic relations between the roots and coefficients of polynomials form a foundational pillar of higher-secondary and undergraduate algebra. This document provides an exhaustive, 101-category taxonomy ranging from fundamental quadratic symmetric functions to advanced analytic, linear algebraic, and topological properties of polynomial roots.

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## 2 Taxonomy of 101 Problem Categories

### 2.1 Basic Quadratic Foundations & Algebraic Identities

#### 2.1.1 Category 1: Basic Evaluation of Symmetric Functions of Roots

Evaluating fundamental symmetric combinations using  $\alpha + \beta = -\frac{b}{a}$  and  $\alpha\beta = \frac{c}{a}$ .

- **Q1:** If  $\alpha, \beta$  are roots of  $x^2 - 4x + 2 = 0$ , evaluate  $\alpha^2 + \beta^2$ ,  $\alpha^3 + \beta^3$ , and  $\frac{1}{\alpha} + \frac{1}{\beta}$ .
- **Q2:** If  $\alpha, \beta$  are roots of  $3x^2 - 6x + 4 = 0$ , evaluate  $\left(\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}\right) + \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$ .

#### 2.1.2 Category 2: Non-Symmetric Expressions Involving Roots

Evaluating non-symmetric functions where swapping  $\alpha$  and  $\beta$  alters the output sign or structure.

- **Q1:** If  $\alpha, \beta$  are roots of  $x^2 - 4x + 2 = 0$ , find the exact value of  $\alpha^2 - \beta^2$ .
- **Q2:** Express  $\alpha^3 - \beta^3$  in terms of the coefficients  $a, b, c$  for  $ax^2 + bx + c = 0$ .

#### 2.1.3 Category 3: Advanced Polynomial & Negative Power Algebraic Identities

Manipulating higher powers and negative exponent combinations of roots.

- **Q1:** If  $\alpha, \beta$  are roots of  $ax^2 + bx + c = 0$ , evaluate  $(a\alpha + b)^{-2} + (a\beta + b)^{-2}$ .
- **Q2:** Find the expression for  $\alpha^6 + \beta^6$  in terms of  $a, b, c$ .

#### 2.1.4 Category 4: Given Difference of Roots

Solving for parameters when  $|\alpha - \beta|$  or  $(\alpha - \beta)^2$  is constrained.

- **Q1:** Find  $p$  such that the roots of  $x^2 - px + 10 = 0$  differ by 3.
- **Q2:** Solve  $x^2 + px + 45 = 0$ , given that  $(\alpha - \beta)^2 = 144$ .

#### 2.1.5 Category 5: Roots in Given Numerical Ratios

Analyzing equations where roots satisfy a given ratio  $\alpha : \beta = p : q$ .

- **Q1:** Derive the condition on  $a, b, c$  for the roots of  $ax^2 + bx + c = 0$  to be in the ratio  $p : q$ .
- **Q2:** If the ratio of the roots of  $x^2 + \alpha x + \alpha + 2 = 0$  is 2, determine  $\alpha$ .

#### 2.1.6 Category 6: One Root is a Multiple/Fraction of the Other

Special case of ratio constraints where  $\beta = n\alpha$ .

- **Q1:** For what values of  $a$  is one root of  $x^2 + (2a + 1)x + a^2 + 2 = 0$  double the other?
- **Q2:** Determine  $k$  if one root of  $k(x - 1)^2 = 5x - 7$  is twice the other.

#### 2.1.7 Category 7: One Root is the Power of the Other

Problems where  $\beta = \alpha^2$  or  $\beta = \alpha^n$ .

- **Q1:** If one root of  $ax^2 + bx + c = 0$  is the square of the other, prove that  $b^3 + ac(c + a) = 3abc$ .
- **Q2:** Find  $a$  if one root of  $8x^2 - 6x + a = 0$  is the square of the other.

### 2.1.8 Category 8: Equal Roots in Magnitude but Opposite in Sign / Reciprocal Roots

Parameter conditions for symmetric cancellation ( $\beta = -\alpha$ ) or reciprocal relation ( $\alpha\beta = 1$ ).

- **Q1:** Find  $k$  such that  $(k-1)x^2 - kx + 1 = 0$  has roots numerically equal but opposite in sign.
- **Q2:** State the condition for one root of  $ax^2 + bx + c = 0$  to be the reciprocal of the other.

### 2.1.9 Category 9: Roots Differing by a Fixed Constant

Cases where  $\beta = \alpha + k$ .

- **Q1:** If the roots of  $x^2 - lx + m = 0$  differ by 1, prove  $l^2 = 4m + 1$ .
- **Q2:** Find  $k$  such that one root of  $2kx^2 - 20x + 21 = 0$  exceeds the other by 2.

### 2.1.10 Category 10: Linear Relations Between Roots

Solving subject to custom linear relations  $A\alpha + B\beta = C$ .

- **Q1:** Find  $k$  if the roots  $\alpha, \beta$  of  $x^2 - 6x + k = 0$  satisfy  $2\alpha + 3\beta = 20$ .
- 

## 2.2 Equation Formation & Root Transformations

### 2.2.1 Category 11: Formation of Equations with Given Numerical/Complex Roots

Constructing polynomials  $x^2 - Sx + P = 0$  given specific roots or conjugate pairs.

- **Q1:** Form a quadratic equation with rational coefficients having one root as  $\frac{1}{3+2\sqrt{2}}$ .
- **Q2:** Form an equation with real coefficients having  $-2 + i$  as a root.

### 2.2.2 Category 12: Formation of Equations with Scaled or Shifted Roots

Constructing equations whose roots are  $n\alpha, n\beta$  or  $\alpha + k, \beta + k$ .

- **Q1:** If  $\alpha, \beta$  are roots of  $ax^2 + bx + c = 0$ , form an equation with roots  $n\alpha, n\beta$ .
- **Q2:** Form an equation whose roots are  $\alpha + k$  and  $\beta + k$ .

### 2.2.3 Category 13: Formation of Equations with Reciprocal / Symmetric Transformations

Transformations such as  $\frac{1}{\alpha}, \frac{1}{\beta}$  or  $\alpha^2, \beta^2$ .

- **Q1:** Form an equation whose roots are  $\frac{1}{\alpha}$  and  $\frac{1}{\beta}$ .
- **Q2:** Form an equation whose roots are  $\alpha^3$  and  $\beta^3$ .

### 2.2.4 Category 14: Formation of Equations with Composite Rational Expressions

New roots defined by multi-term rational combinations like  $\frac{\alpha}{2\beta+3}$ .

- **Q1:** If  $\alpha, \beta$  are roots of  $2x^2 - 3x + 1 = 0$ , form an equation with roots  $\frac{\alpha}{2\beta+3}$  and  $\frac{\beta}{2\alpha+3}$ .
  - **Q2:** Form an equation with roots  $\alpha + \frac{1}{\beta}$  and  $\beta + \frac{1}{\alpha}$ .
- 

## 2.3 Multi-Polynomial Intersections & Common Roots

### 2.3.1 Category 15: Single Common Root Between Two Equations

Conditions for two quadratics to share exactly one root.

- **Q1:** Find  $k$  such that  $3x^2 + kx + 2 = 0$  and  $2x^2 + 3x - 2 = 0$  share a common root.
- **Q2:** Prove  $a + b + 1 = 0$  if  $x^2 + ax + b = 0$  and  $x^2 + bx + a = 0$  share a root ( $a \neq b$ ).

### 2.3.2 Category 16: Both Roots Common Between Two Quadratic Equations

Condition for two equations to have identical root sets ( $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ ).

- **Q1:** State the condition for  $a_1x^2 + b_1x + c_1 = 0$  and  $a_2x^2 + b_2x + c_2 = 0$  to have both roots common.

### 2.3.3 Category 17: Equating Difference of Roots Across Two Different Equations

Parameter relations when  $|\alpha_1 - \beta_1| = |\alpha_2 - \beta_2|$ .

- **Q1:** If the root differences of  $x^2 + px + q = 0$  and  $x^2 + qx + p = 0$  are equal, prove  $p + q + 4 = 0$  ( $p \neq q$ ).

### 2.3.4 Category 18: Reconstructing Equations from Incorrectly Copied Coefficients

Deducing correct roots when individual terms are misread.

- **Q1:** Student A miscopies  $p$  and gets roots 2, 6. Student B miscopies  $q$  and gets roots 2, -9. Find the correct roots.
- **Q2:** The linear coefficient  $p$  was misread as 17 instead of 13, yielding roots -2, -15. Find original roots.

### 2.3.5 Category 19: Geometric Progressions of Roots Across Two Equations

Sequence coupling across roots of two distinct quadratics ( $\alpha, \beta, \gamma, \delta \in \text{G.P.}$ ).

- **Q1:** If roots of  $x^2 + px + q = 0$  and  $x^2 + bx + c = 0$  form a continuous G.P., prove  $p^2c = b^2q$ .

### 2.3.6 Category 20: Conditions for Roots to Have Specific Signs

Determining parameter ranges for positive, negative, or opposite-sign roots.

- **Q1:** Calculate  $a$  if both roots of  $x^2 - (a + 1)x + a + 4 = 0$  are negative.
- **Q2:** If roots of  $2x^2 + (k + 1)x + (k^2 - 5k + 6) = 0$  are of opposite signs, show  $2 < k < 3$ .

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## 2.4 Degree-N Systems & Higher Order Vieta Relations

### 2.4.1 Category 21: Substituting Roots Back into Quadratic Polynomials

Simplification using  $a\alpha^2 + b\alpha + c = 0 \implies a\alpha + b = -\frac{c}{\alpha}$ .

- **Q1:** If  $\alpha, \beta$  are roots of  $x^2 = 7x + 4$ , show  $\alpha^3 = 53\alpha + 28$ .
- **Q2:** Form an equation with roots  $\frac{1}{p\alpha - q}$  and  $\frac{1}{p\beta - q}$  for  $x^2 - px + q = 0$ .

### 2.4.2 Category 22: Cubic Equations – Basic Symmetric Sums of Roots

Evaluating  $\sum \alpha, \sum \alpha\beta, \alpha\beta\gamma$  for cubic polynomials.

- **Q1:** If  $\alpha, \beta, \gamma$  are roots of  $x^3 - 6x^2 + 11x - 6 = 0$ , evaluate  $\alpha^2 + \beta^2 + \gamma^2$ .
- **Q2:** For  $2x^3 - 3x^2 + 5x - 1 = 0$ , evaluate  $\alpha^3 + \beta^3 + \gamma^3$ .

### 2.4.3 Category 23: Higher Degree Polynomials (Biquadratic & Degree $n$ )

Vieta's relations applied to polynomials of degree 4 or higher.

- **Q1:** For  $x^4 - 4x^3 + 8x^2 - 7x + 3 = 0$ , evaluate  $\sum \alpha^2\beta$ .
- **Q2:** Find the product of roots of  $3x^5 - 15x^4 + 2x^2 - 9 = 0$ .

### 2.4.4 Category 24: Roots of Polynomials in Arithmetic Progression (A.P.)

Solving polynomials whose roots form an A.P. ( $\alpha - d, \alpha, \alpha + d$ ).

- **Q1:** Solve  $x^3 - 12x^2 + 39x - 28 = 0$  given its roots are in A.P.
- **Q2:** Derive the coefficient condition for  $ax^3 + bx^2 + cx + d = 0$  to have roots in A.P.

### 2.4.5 Category 25: Roots of Polynomials in Harmonic Progression (H.P.)

Solving polynomials with roots in H.P. via substitution  $x = 1/y$ .

- **Q1:** Solve  $6x^3 - 11x^2 + 6x - 1 = 0$  given its roots are in H.P.
- **Q2:** Find the condition on coefficients of  $x^3 - px^2 + qx - r = 0$  for roots to be in H.P.

### 2.4.6 Category 26: Trigonometric Relations Between Roots

Parametrization of roots as trigonometric values  $(\tan A, \tan B)$ .

- **Q1:** If  $\tan A, \tan B$  are roots of  $x^2 - px + q = 0$ , express  $\sin^2(A + B)$  in terms of  $p, q$ .
- **Q2:** If  $\cos \theta_1, \cos \theta_2$  are roots of  $ax^2 + bcx + c = 0$ , evaluate  $\cos(\theta_1 + \theta_2)$ .

### 2.4.7 Category 27: Newton's Sums / Newton's Theorem on Powers of Roots

Utilizing  $aS_n + bS_{n-1} + cS_{n-2} = 0$  for efficient power-sum computation.

- **Q1:** If  $\alpha, \beta$  are roots of  $x^2 - 6x - 2 = 0$  and  $a_n = \alpha^n - \beta^n$ , evaluate  $\frac{a_{10} - 2a_8}{2a_9}$ .
- **Q2:** For  $x^2 - x - 1 = 0$ , prove  $P_{n+1} = P_n + P_{n-1}$  where  $P_n = \alpha^n + \beta^n$ .

### 2.4.8 Category 28: Recurrence Relations for Symmetric Power Sums

Formulating linear recurrence relations for  $S_n = \alpha^n + \beta^n$ .

- **Q1:** Express  $S_n$  in terms of  $S_{n-1}$  and  $S_{n-2}$  for  $x^2 - 3x + 1 = 0$ , and evaluate  $\alpha^5 + \beta^5$ .

### 2.4.9 Category 29: Common Roots Between Quadratic and Higher Degree Equations

Intersection of roots between quadratics and cubics/quartic polynomials.

- **Q1:** If  $x^2 - 3x + 2 = 0$  and  $x^3 - 6x^2 + ax - 6 = 0$  share a root, find  $a$ .
- **Q2:** Find  $k$  such that  $x^2 + kx - 6 = 0$  and  $x^3 - 2x^2 - x + 2 = 0$  share a root.

### 2.4.10 Category 30: Equations with Imaginary / Complex Coefficients

Vieta's relations extended to polynomials with complex coefficients  $a, b, c \in \mathbb{C}$ .

- **Q1:** For  $x^2 - (3 + 2i)x + (1 + 3i) = 0$ , form an equation with roots  $2\alpha, 2\beta$ .
- **Q2:** Find the condition on  $k \in \mathbb{C}$  for  $x^2 + ix + k = 0$  to have one real root.

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## 2.5 Transformations, Modular Systems, & Bounds

### 2.5.1 Category 31: Transformation of Equations – Diminishing or Increasing Roots

Constructing equations with shifted roots  $\alpha - h, \beta - h$ .

- **Q1:** Transform  $x^3 - 6x^2 + 11x - 6 = 0$  into an equation whose roots are diminished by 2.
- **Q2:** Find the equation whose roots are greater by 3 than the roots of  $2x^2 - 5x + 1 = 0$ .

### 2.5.2 Category 32: Transformation of Equations – Reciprocating Higher Degree Roots

Inverting roots for polynomials of degree  $n \geq 3$ .

- **Q1:** Form the equation whose roots are reciprocals of the roots of  $ax^3 + bx^2 + cx + d = 0$ .
- **Q2:** For  $x^3 - 2x^2 + 3x - 4 = 0$ , find the equation with roots  $\frac{1}{\alpha\beta}, \frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}$ .

### 2.5.3 Category 33: Integral / Rational Root Constraints

Combining Vieta's formulas with integer divisibility  $(\alpha, \beta \in \mathbb{Z})$ .

- **Q1:** Find all integral values of  $p$  for which  $x^2 - px + 6 = 0$  has integer roots.
- **Q2:** If  $x^2 + ax + b = 0$  has rational roots  $(a, b \in \mathbb{Z})$ , prove  $a^2 - 4b$  is a perfect square.

#### 2.5.4 Category 34: Condition for Negative Reciprocal Roots ( $\alpha\beta = -1$ )

Parameter conditions when one root is the negative reciprocal of the other.

- **Q1:** Derive the coefficient condition for  $ax^2 + bx + c = 0$  to satisfy  $\alpha\beta = -1$ .
- **Q2:** Find  $k$  such that  $2x^2 + 5x + (k - 3) = 0$  has negative reciprocal roots.

#### 2.5.5 Category 35: Systems of Non-Linear Symmetric Equations Solved via Roots

Solving symmetric system of equations by constructing auxiliary polynomials.

- **Q1:** Solve for  $x, y$ :  $x + y = 7, x^3 + y^3 = 91$ .
- **Q2:** Solve for real  $x, y, z$ :  $x + y + z = 6, xy + yz + zx = 11, xyz = 6$ .

#### 2.5.6 Category 36: Logarithmic and Exponential Expressions Involving Roots

Evaluating logarithmic or exponential combinations of roots.

- **Q1:** If  $\alpha, \beta > 0$  are roots of  $x^2 - 10x + 16 = 0$ , evaluate  $\log_2 \alpha + \log_2 \beta$ .
- **Q2:** For  $x^2 - 5x + 2 = 0$ , evaluate  $e^{\alpha+\beta} \cdot e^{-\alpha\beta}$ .

#### 2.5.7 Category 37: Geometric Progression of Roots in Higher Degree Equations

Solving polynomials whose roots form a G.P. ( $\frac{a}{r}, a, ar$ ).

- **Q1:** Solve  $x^3 - 7x^2 + 14x - 8 = 0$  given its roots are in G.P.
- **Q2:** Find coefficient relations for  $ax^3 + bx^2 + cx + d = 0$  to have roots in G.P.

#### 2.5.8 Category 38: Sum of Two Roots Equals Zero ( $\beta = -\alpha$ ) in Higher Degrees

Symmetric cancellation in cubic or quartic equations.

- **Q1:** Solve  $x^3 - 3x^2 - x + 3 = 0$  given two roots sum to zero.
- **Q2:** Find the condition on  $p, q, r$  for  $x^3 + px^2 + qx + r = 0$  to have two roots summing to zero.

#### 2.5.9 Category 39: Product of Two Roots Equals Unity ( $\alpha\beta = 1$ ) in Higher Degrees

Higher degree polynomials with a pair of reciprocal roots.

- **Q1:** Solve  $2x^3 - 9x^2 + 12x - 4 = 0$  given two roots multiply to 1.
- **Q2:** Derive the condition for  $ax^3 + bx^2 + cx + d = 0$  to have two roots whose product is 1.

#### 2.5.10 Category 40: One Root is the Sum of Other Roots ( $\gamma = \alpha + \beta$ )

Polynomials where one root equals the sum of all remaining roots.

- **Q1:** Solve  $x^3 - 6x^2 + 11x - 6 = 0$  given one root equals the sum of the other two.

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### 2.6 Reciprocal Equations, Geometry, & inequalities

#### 2.6.1 Category 41: Biquadratic Equations with Roots in A.P. / G.P.

Solving degree 4 polynomials with structured root progressions.

- **Q1:** Solve  $x^4 - 10x^2 + 9 = 0$  given its roots are in A.P.
- **Q2:** Find  $k$  such that the roots of  $x^4 - 15x^2 + k = 0$  are in A.P.

#### 2.6.2 Category 42: Symmetric / Reciprocal Equations of Degree $n$

Palindromic polynomials ( $a_k = a_{n-k}$ ) invariant under  $x \rightarrow 1/x$ .

- **Q1:** Solve  $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$ .
- **Q2:** Show that if  $\alpha$  is a root of  $ax^4 + bx^3 + cx^2 + bx + a = 0$ , then  $1/\alpha$  is also a root.



### 2.6.3 Category 43: Roots as Vertices of Geometric Shapes in Complex Plane

Connecting roots to Argand plane geometric configurations (equilateral triangles, squares).

- **Q1:** If roots of  $z^3 + pz^2 + qz + r = 0$  form an equilateral triangle in  $\mathbb{C}$ , prove  $p^2 = 3q$ .
- **Q2:** If roots of  $z^2 + az + b = 0$  and the origin form an equilateral triangle, prove  $a^2 = 3b$ .

### 2.6.4 Category 44: Trigonometric Products and Sums via Polynomial Roots

Constructing polynomials with roots like  $\tan^2(\pi/7)$  to derive exact trigonometric sums.

- **Q1:** Form a cubic with roots  $\tan^2(\frac{\pi}{7}), \tan^2(\frac{2\pi}{7}), \tan^2(\frac{3\pi}{7})$  and find their sum.
- **Q2:** Evaluate  $\cos(\frac{\pi}{5}) \cos(\frac{2\pi}{5})$  using roots of a quadratic.

### 2.6.5 Category 45: Vector/Dot Product Parameterization of Roots

Connecting root values to vector magnitudes and dot products.

- **Q1:** If roots  $\alpha, \beta$  of  $x^2 - 2px + q = 0$  represent vector magnitudes  $|\vec{a}|, |\vec{b}|$ , express  $\theta = \angle(\vec{a}, \vec{b})$  in terms of  $p, q, |\vec{a} + \vec{b}|$ .

### 2.6.6 Category 46: Functional Equations Involving Roots

Evaluating expressions of the form  $f(\alpha) + f(\beta)$  for a given function  $f(x)$ .

- **Q1:** Evaluate  $f(\alpha) + f(\beta)$  for  $f(x) = x^3 - 2x^2 + 5x + 7$  where  $\alpha, \beta$  are roots of  $x^2 - 2x + 5 = 0$ .
- **Q2:** For  $g(x) = \frac{x^2+1}{x}$  and roots of  $x^2 - x - 1 = 0$ , find  $g(\alpha)g(\beta)$ .

### 2.6.7 Category 47: Sums of Squares of Differences $(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2$

Symmetric dispersion metrics for cubic and higher-order roots.

- **Q1:** For  $x^3 + px + q = 0$ , evaluate  $(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2$  in terms of  $p, q$ .

### 2.6.8 Category 48: Transforming Roots to Squares $(\alpha^2, \beta^2, \gamma^2)$

Constructing equations satisfied by squared roots of higher-degree polynomials.

- **Q1:** Form an equation whose roots are  $\alpha^2, \beta^2, \gamma^2$  for  $x^3 - px^2 + qx - r = 0$ .

### 2.6.9 Category 49: Equations with Complex Conjugate Pair Constraints

Utilizing  $\beta = \bar{\alpha}$  for real-coefficient polynomials to find remaining roots.

- **Q1:** Given  $2 + 3i$  is a root of  $x^3 - 7x^2 + 19x - 13 = 0$ , find the real root and verify the root product.

### 2.6.10 Category 50: Irrational Conjugate Pair Constraints $(p + \sqrt{q})$

Properties of irrational conjugate pairs in polynomials with rational coefficients.

- **Q1:** If  $1 + \sqrt{2}$  is a root of  $x^4 - 2x^3 - 3x^2 + 2x + 1 = 0$ , find all roots and the sum of their squares.

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## 2.7 Analytic Properties, Derivatives, & Advanced Applications

### 2.7.1 Category 51: Bounds on Roots via Vieta's Identities

Applying inequalities (AM-GM, Cauchy-Schwarz) to root-coefficient relations.

- **Q1:** If all roots of  $x^3 - 6x^2 + px - 8 = 0$  are positive real numbers, find  $p$  using the AM-GM inequality.
- **Q2:** Prove  $\alpha^2 + \beta^2 \geq \frac{b^2}{2}$  for real roots of  $x^2 + bx + c = 0$ .

### 2.7.2 Category 52: Determinant / Matrix Expressions of Root Systems

Evaluating determinants whose components consist of polynomial roots.

- **Q1:** Evaluate  $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$  where  $\alpha, \beta, \gamma$  are roots of  $x^3 + px + q = 0$ .

### 2.7.3 Category 53: Modular Arithmetic / Congruence Relations on Root Sums

Evaluating power-sum values modulo  $m$  ( $S_n \pmod{m}$ ).

- **Q1:** Prove  $\alpha^n + \beta^n \in \mathbb{Z}$  for roots of  $x^2 - 4x + 1 = 0$ , and evaluate its value modulo 3.

### 2.7.4 Category 54: Calculus / Derivative Relations with Root Sums

Connecting derivatives to root reciprocals:  $\frac{f'(x)}{f(x)} = \sum \frac{1}{x - \alpha_i}$ .

- **Q1:** For  $f(x) = x^3 - 5x + 3$ , evaluate  $\frac{1}{f'(\alpha)} + \frac{1}{f'(\beta)} + \frac{1}{f'(\gamma)}$ .
- **Q2:** Express  $\frac{1}{\alpha - x} + \frac{1}{\beta - x}$  in terms of  $f(x)$  and  $f'(x)$  for  $f(x) = ax^2 + bx + c$ .

### 2.7.5 Category 55: Root Continuity under Parameter Perturbations

Estimating variations in root sums/products under coefficient shifts ( $c \rightarrow c + \epsilon$ ).

- **Q1:** Find the linear approximation for the change in root sum and product when  $x^2 - 5x + 6 = 0$  becomes  $x^2 - 5x + (6 + \epsilon) = 0$ .

### 2.7.6 Category 56: Generating Functions for Power Sums of Roots

Formal power series  $G(t) = \sum_{n=0}^{\infty} S_n t^n$  derived from Vieta's identities.

- **Q1:** Find the generating function for  $S_n = \alpha^n + \beta^n$  for  $x^2 - 3x + 2 = 0$ , and evaluate  $S_6$ .

### 2.7.7 Category 57: Equations Solvable via Auxiliary Substitution of Root Products

Simplifying systems by setting  $y = \alpha\beta$  or  $y = \alpha + \beta$ .

- **Q1:** If  $\alpha^2 + \beta^2 = 5$  for  $x^2 - px + q = 0$ , express  $p$  in terms of  $q$ .
- **Q2:** Find  $k$  if roots of  $x^2 - 8x + k = 0$  satisfy  $\alpha^2 + \beta^2 = 40$ .

### 2.7.8 Category 58: Roots of Unity and Cyclotomic Polynomial Relations

Vieta's relations applied to cyclotomic equations  $x^n - 1 = 0$ .

- **Q1:** For roots  $1, \omega, \omega^2$  of  $x^3 - 1 = 0$ , evaluate  $(1 - \omega + \omega^2)(1 + \omega - \omega^2)$ .
- **Q2:** Compute  $\prod_{k=1}^{n-1} (1 - e^{\frac{2\pi i k}{n}})$  using relations between roots and coefficients.

### 2.7.9 Category 59: Symmetric Functions Involving Fractional Powers ( $\sqrt{\alpha} + \sqrt{\beta}$ )

Radical expressions of roots derived from elementary symmetric functions.

- **Q1:** Evaluate  $\sqrt{\alpha} + \sqrt{\beta}$  for positive roots of  $x^2 - 10x + 9 = 0$ .
- **Q2:** Express  $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}}$  in terms of  $a, b, c$  for  $ax^2 + bx + c = 0$ .

### 2.7.10 Category 60: Quadratic Forms and Discriminant Invariants

Direct identification of  $(\alpha - \beta)^2 = \frac{D}{a^2}$  with discriminant invariants.

- **Q1:** Prove  $(\alpha - \beta)^2 = \frac{D}{a^2}$  for  $ax^2 + bx + c = 0$  with discriminant  $D = b^2 - 4ac$ .
- **Q2:** Express  $\alpha^4 + \beta^4$  in terms of  $a, c$  and discriminant  $D$ .

### 2.7.11 Category 61: Roots in Harmonic Progression (H.P.) for Quadratics

Arithmetic constraints on reciprocal roots  $1/\alpha, 1/\beta$ .

- **Q1:** If  $1/\alpha, 1/\beta$  are in A.P. with common difference  $d$ , express  $d^2$  in terms of coefficients of  $ax^2 + bx + c = 0$ .

### 2.7.12 Category 62: Sums of Cubes of Root Differences $(\alpha - \beta)^3 + (\beta - \gamma)^3 + (\gamma - \alpha)^3$

Cyclic cubic identities applied to root differences.

- **Q1:** Prove  $(\alpha - \beta)^3 + (\beta - \gamma)^3 + (\gamma - \alpha)^3 = 3(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$  for  $x^3 + px + q = 0$ .

### 2.7.13 Category 63: Maximum and Minimum Values of Symmetric Functions

Optimization of symmetric root functions subject to parameter constraints.

- **Q1:** Find the minimum value of  $\alpha^2 + \beta^2$  for real roots of  $x^2 - 2px + (p^2 - 1) = 0$ .
- **Q2:** Find the maximum value of  $\alpha\beta(\alpha + \beta)$  for roots of  $x^2 - mx + 4 = 0$  ( $m \in [4, 6]$ ).

### 2.7.14 Category 64: Equations with Roots as Specific Geometric Parameters

Relating roots to geometric measurements (triangle sides, inradius, circumradius).

- **Q1:** If roots of  $x^2 - 7x + 12 = 0$  represent leg lengths of a right triangle, find its hypotenuse.
- **Q2:** Roots of  $x^2 - 2Rx + A = 0$  represent circumradius  $R$  and inradius  $r$ . Express the distance between circumcenter and incenter using roots.

### 2.7.15 Category 65: Operator/Differential Invariants on Polynomial Roots

Utilizing higher order logarithmic derivative operators  $\frac{d}{dx} \left( \frac{f'(x)}{f(x)} \right) = - \sum \frac{1}{(x - \alpha_i)^2}$ .

- **Q1:** Evaluate  $\frac{1}{(1-\alpha)^2} + \frac{1}{(1-\beta)^2} + \frac{1}{(1-\gamma)^2}$  for roots of  $x^3 - x - 1 = 0$ .

### 2.7.16 Category 66: System of Equations with Alternating Signs of Roots

Combining Descartes' Rule of Signs with Vieta's formulas.

- **Q1:** If  $x^3 - 4x^2 + x + 6 = 0$  has two positive roots and one negative root, find the product of the positive roots.

### 2.7.17 Category 67: Roots of Polynomials over Finite Fields ( $\mathbb{Z}_p$ )

Evaluating root relations under modular arithmetic field structures.

- **Q1:** Find the sum and product of roots of  $x^2 + 3x + 2 \equiv 0 \pmod{7}$ .
- **Q2:** Determine if  $x^2 + x + 1 \equiv 0 \pmod{5}$  has roots, and evaluate their sum in  $\mathbb{Z}_5$ .

### 2.7.18 Category 68: Transformation via Logarithmic Differentiation

Evaluating rational sums  $\sum \frac{1}{k - \alpha_i}$  at fixed evaluation points.

- **Q1:** Evaluate  $\sum_{i=1}^4 \frac{1}{2 - \alpha_i}$  for roots of  $x^4 - 3x^3 + 2x^2 - x + 5 = 0$ .

### 2.7.19 Category 69: Symmetric Sums of Inverse Cubes $\sum \frac{1}{\alpha^3}$

Evaluating inverse cubic power sums via reciprocal polynomial transformations.

- **Q1:** Evaluate  $\frac{1}{\alpha^3} + \frac{1}{\beta^3} + \frac{1}{\gamma^3}$  for roots of  $x^3 - 2x^2 + 4x - 8 = 0$ .

### 2.7.20 Category 70: Roots of Multi-Variable Homogeneous Quadratic Forms

Slopes of binary quadratic forms  $ax^2 + bxy + cy^2 = 0$  as roots of  $am^2 + bm + c = 0$ .

- **Q1:** If  $m_1, m_2$  are slopes represented by  $3x^2 - 8xy + 4y^2 = 0$ , evaluate  $m_1 + m_2$  and  $m_1 m_2$ .

## 2.8 Specialized Structure, Physics, & Special Functions

### 2.8.1 Category 71: Sums of Roots Taking $k$ at a Time ( $\sigma_k$ ) for Degree $n$

Direct extraction of intermediate elementary symmetric functions  $\sigma_k$ .

- **Q1:** Extract  $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5$  for  $x^5 - 5x^4 + 10x^3 - 10x^2 + 5x - 1 = 0$ .

### 2.8.2 Category 72: Characterizing Equations with Purely Imaginary Roots

Coefficient constraints ensuring  $\operatorname{Re}(\alpha) = 0$  for all roots.

- **Q1:** Find conditions on  $b, c \in \mathbb{R}$  for  $x^2 + bx + c = 0$  to have purely imaginary roots.
- **Q2:** Prove  $x^3 + bx^2 + cx + d = 0$  cannot have purely imaginary roots if  $b \neq 0$ .

### 2.8.3 Category 73: Parameter Elimination in Coupled Root Equations

Eliminating parametric variables  $t$  to establish direct Cartesian relations.

- **Q1:** Eliminate  $t$  from  $x^2 - 2tx + (t^2 - 1) = 0$  to find a relation between  $\alpha + \beta$  and  $\alpha\beta$ .

### 2.8.4 Category 74: Polynomials with Roots in Exponential Progression

Roots forming exponential terms  $e^a, e^b, e^c$  with exponents in A.P.

- **Q1:** If roots of  $x^3 - px^2 + qx - r = 0$  are  $e^a, e^b, e^c$  with  $a, b, c \in \text{A.P.}$ , prove  $q^3 = p^3r$ .

### 2.8.5 Category 75: Continued Fraction Representations Derived from Roots

Expressing roots as infinite continued fractions  $\alpha = x_1 + \frac{1}{\alpha}$ .

- **Q1:** Express the positive root of  $x^2 - x - 1 = 0$  as  $1 + \frac{1}{1 + \frac{1}{1 + \dots}}$ , and evaluate using Vieta's relations.

### 2.8.6 Category 76: Hyperbolic Function Parameterization of Roots

Utilizing  $\cosh t, \sinh t$  in cubic evaluation.

- **Q1:** If  $\cosh t, \sinh t$  are roots of  $x^2 - px + q = 0$ , express the relation between  $p, q$  using hyperbolic identities.

### 2.8.7 Category 77: Roots Involving Floor / Ceiling Functions

Tying integer bounds  $\lfloor \alpha \rfloor, \lceil \beta \rceil$  to polynomial coefficients.

- **Q1:** For roots of  $x^2 - 7x + 11 = 0$ , evaluate  $\lfloor \alpha \rfloor + \lceil \beta \rceil$  and  $\lfloor \alpha + \beta \rfloor$ .
- **Q2:** Find integral  $k$  values such that roots of  $x^2 - kx + 6 = 0$  satisfy  $\lfloor \alpha \rfloor = \lfloor \beta \rfloor$ .

### 2.8.8 Category 78: Roots Related by Complex Phase / Argument Differences

Constraints on phase angle differences  $|\arg(\alpha) - \arg(\beta)|$  in the Argand plane.

- **Q1:** If roots of  $z^2 + az + b = 0$  have equal magnitudes and phase difference  $\pi/2$ , prove  $a^2 = 2b$ .
- **Q2:** Find condition on  $p, q$  for roots of  $z^2 + pz + q = 0$  to lie on  $\arg(z) = \pi/4$ .

### 2.8.9 Category 79: Symmetric Power Sums with Alternating Signs

Alternating power sum combinations  $\alpha^3 - \beta^3 + \gamma^3$  for ordered roots.

- **Q1:** Evaluate  $\alpha^3 - \beta^3 + \gamma^3$  for roots of  $x^3 - 2x^2 - x + 2 = 0$  ordered as  $\alpha < \beta < \gamma$ .

### 2.8.10 Category 80: Roots Parameterized by Hyperbolic Substitution ( $x = \cosh t$ )

Solving  $4x^3 - 3x - a = 0$  ( $|a| > 1$ ) using hyperbolic triple-angle identities.

- **Q1:** Use  $\cosh(3t) = 4\cosh^3 t - 3\cosh t$  to find real roots of  $4x^3 - 3x - 2 = 0$ .

### 2.8.11 Category 81: Self-Conjugate Root Sets under Möbius Inversion

Invariance of root sets under rational transformations  $x \rightarrow \frac{1-x}{1+x}$ .

- **Q1:** Prove that if  $\alpha$  is a root of  $x^3 + x^2 - 2x - 1 = 0$ , then  $\beta = 2 - \alpha^2$  is also a root.

### 2.8.12 Category 82: Sums of Weighted Powers of Roots ( $\lambda\alpha^n + \mu\beta^n$ )

Constructing non-symmetric sequences using initial values and root recurrences.

- **Q1:** Find  $A, B$  such that  $L_n = A\alpha^n + B\beta^n$  satisfies  $L_1 = 1, L_2 = 3$  for roots of  $x^2 - x - 1 = 0$ .

### 2.8.13 Category 83: Orthogonal Polynomial Root Relations (Chebyshev / Legendre)

Deriving trigonometric identities from roots of Chebyshev polynomials  $T_n(x)$ .

- **Q1:** Evaluate the root sum and product of  $T_3(x) = 4x^3 - 3x = 0$ , verifying connection to  $\cos(\pi/6), \cos(\pi/2), \cos(5\pi/6)$ .

### 2.8.14 Category 84: Polynomials with Integer Coefficients Having All Real Roots

Applying Rolle's theorem and Vieta's relations simultaneously to derive coefficient bounds.

- **Q1:** If all roots of  $x^3 - 6x^2 + px - 8 = 0$  are real, prove  $p = 12$  using  $f'(x) = 0$ .

### 2.8.15 Category 85: Symmetric Functions Involving Binomial Coefficients

Evaluating combinatorial expressions  $\binom{\alpha}{2} + \binom{\beta}{2}$  evaluated at polynomial roots.

- **Q1:** Evaluate  $\binom{\alpha}{2} + \binom{\beta}{2}$  for roots of  $x^2 - 9x + 20 = 0$ .
- **Q2:** Express  $\binom{\alpha}{3} + \binom{\beta}{3}$  in terms of  $p, q$  for roots of  $x^2 - px + q = 0$ .

### 2.8.16 Category 86: Equations with Roots as Eigenvalues of Matrices

Equating matrix trace and determinant to root sums and products:  $\det(A - \lambda I) = 0$ .

- **Q1:** For  $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ , form the characteristic equation and evaluate  $\lambda_1^2 + \lambda_2^2$ .

### 2.8.17 Category 87: Vieta's Formulas Applied to Infinite Polynomial Series

Extending product expansions to transcendental entire functions ( $\sin x = x \prod (1 - x^2/n^2\pi^2)$ ).

- **Q1:** Evaluate the sum of reciprocals of squared non-zero roots of  $\sin x = 0$ .

### 2.8.18 Category 88: Cyclic Expressions of the Form $(\alpha + \omega\beta + \omega^2\gamma)^3$

Utilizing Lagrange resolvents and complex cube roots of unity  $\omega$ .

- **Q1:** Express  $L = \alpha + \omega\beta + \omega^2\gamma$  in terms of  $p, q$  for  $x^3 - 3px + q = 0$ .

### 2.8.19 Category 89: Resultants and Elimination via Root Products

Determining root-sharing conditions via the polynomial resultant  $R(f, g) = a^m b^n \prod (\alpha_i - \beta_j)$ .

- **Q1:** Compute the resultant of  $f(x) = x^2 - px + q$  and  $g(x) = x - k$ .

### 2.8.20 Category 90: Roots of Polynomials under Quadratic Transformations

Constructing equations satisfied by  $y = ax^2 + bx + c$  when  $x$  satisfies a given cubic equation.

- **Q1:** If  $\alpha, \beta, \gamma$  are roots of  $x^3 - x - 1 = 0$ , find the cubic equation with roots  $y_i = \alpha_i^2 + 1$ .

### 2.8.21 Category 91: Sums of Cross-Ratios of Four Roots

Evaluating projective geometric invariants  $(\alpha, \beta; \gamma, \delta) = \frac{(\alpha-\gamma)(\beta-\delta)}{(\alpha-\delta)(\beta-\gamma)}$ .

- **Q1:** Evaluate the sum of all cross-ratios formed by the roots of  $x^4 + px + q = 0$ .

### 2.8.22 Category 92: Equations where Roots Represent Critical Points of Potentials

Physical equilibrium problems where electrostatic net force vanishes ( $\sum \frac{1}{x-\alpha_i} = 0$ ).

- **Q1:** Find the equilibrium point on  $\mathbb{R}$  where net force vanishes between two unit charges placed at roots of  $x^2 - 6x + 8 = 0$ .

### 2.8.23 Category 93: Bounds on Maximum Root Absolute Value ( $|\alpha|_{\max}$ )

Applying Cauchy's bound and Eneström-Kakeya theorem to coefficient magnitudes.

- **Q1:** Show that all roots of  $x^3 + 3x^2 + 4x + 12 = 0$  satisfy  $|\alpha| \leq 13$ , and refine the bound using root products.

### 2.8.24 Category 94: Sum of Roots of Exponential Polynomials

Transforming transcendental equations  $ae^{2x} + be^x + c = 0$  into quadratics  $u = e^x$ .

- **Q1:** Solve  $e^{2x} - 5e^x + 6 = 0$  and find the exact sum of real solutions using the root product of the transformed equation.
- **Q2:** Prove  $x_1 + x_2 = \log_2(c/a)$  for real roots of  $a4^x + b2^x + c = 0$ .

### 2.8.25 Category 95: Trigonometric Sums via Roots of $\cos(n\theta)$ Polynomials

Deriving sum identities  $\sum \cos(\frac{(2k-1)\pi}{2n}) = 0$  using Chebyshev roots.

- **Q1:** Form the polynomial with roots  $\cos(\pi/6), \cos(3\pi/6), \cos(5\pi/6)$  and verify their product.

### 2.8.26 Category 96: Conditions for Roots to Lie Inside the Unit Circle ( $|\alpha| < 1$ )

Applying Schur-Cohn stability criteria to root-coefficient relationships.

- **Q1:** Find the range of  $k$  such that both roots of  $x^2 + kx + 1/2 = 0$  lie strictly inside the unit circle.

### 2.8.27 Category 97: Relations between Roots of a Polynomial and its Antiderivative

Root relationships between  $f(x)$  and  $F(x) = \int_0^x f(t)dt$ .

- **Q1:** For  $f(x) = 3x^2 - 6x + 2$ , find the roots of  $F(x) = \int_0^x f(t)dt$  and express the product of non-zero roots using  $f(x)$  coefficients.

### 2.8.28 Category 98: Distance Between Roots in the Argand Plane ( $|\alpha - \beta|$ )

Computing Euclidean distance between complex conjugate roots  $x = u \pm iv$  using discriminant  $D < 0$ .

- **Q1:** Compute the Euclidean distance in  $\mathbb{C}$  between the conjugate roots of  $x^2 - 4x + 13 = 0$ .

### 2.8.29 Category 99: Systems Solved by Symmetric Polynomial Inversion

Solving non-linear algebraic systems by creating auxiliary polynomials  $t^3 - e_1t^2 + e_2t - e_3 = 0$ .

- **Q1:** Solve  $x + y + z = 2, x^2 + y^2 + z^2 = 14, x^3 + y^3 + z^3 = 20$  by constructing the corresponding cubic equation.

### 2.8.30 Category 100: Sums of Reciprocals of Pairwise Root Sums ( $\sum \frac{1}{\alpha+\beta}$ )

Evaluating symmetric functions formed by adding pairs of roots for cubic equations.

- **Q1:** For  $x^3 - px^2 + qx - r = 0$ , evaluate  $\frac{1}{\alpha+\beta} + \frac{1}{\beta+\gamma} + \frac{1}{\gamma+\alpha}$  in terms of  $p, q, r$ .

### 2.8.31 Category 101: Discriminant of the Cubic Equation via Root Differences

Connecting  $\Delta = a^4(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2$  directly to coefficients of  $x^3 + px + q = 0$ .

- **Q1:** Prove that for  $x^3 + px + q = 0$ , the product of squared root differences  $(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2$  equals  $-4p^3 - 27q^2$ .

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### 3 Top 10 Reference Books

The following ten standard text references cover the theoretical foundations, historical developments, competitive examination frameworks, and advanced problem-solving techniques associated with the 101 categories presented above:

1. **Hall, H. S., & Knight, S. R.** (1887). *Higher Algebra: A Sequel to Elementary Algebra for Schools*. Macmillan and Co. (*The classical benchmark text for Vieta's formulas, symmetric functions, transformations of equations, and reciprocal equations*).
2. **Burnside, W. S., & Panton, A. W.** (1912). *The Theory of Equations: With an Introduction to the Theory of Binary Algebraic Forms* (Vol. 1 & 2). Longmans, Green, and Co. (*An exhaustive historical treatise detailing symmetric functions, Newton's sums, transformed equations, and resultants*).
3. **Uspensky, J. V.** (1948). *Theory of Equations*. McGraw-Hill Book Company. (*Provides rigorous analytical proofs for root bounds, Rolle's applications, symmetric functions, and numerical approximation methods*).
4. **Borwein, P., & Erdélyi, T.** (1995). *Polynomials and Polynomial Inequalities*. Springer-Verlag. (*Covers advanced analytical properties, root distributions, bounds inside the unit circle, and extremal problems*).
5. **Prasolov, V. V.** (2004). *Polynomials* (Algorithms and Computation in Mathematics, Vol. 11). Springer. (*Comprehensive modern theoretical text exploring resultants, symmetric functions, Newton's sums, and complex root geometry*).
6. **Andreescu, T., & Gelca, R.** (2017). *Mathematical Olympiad Challenges*. Birkhäuser / Springer. (*Focuses on high-level problem-solving involving symmetric polynomials, matrix eigenvalue links, and complex root configurations*).
7. **Agarwal, R. S.** (2020). *Senior Secondary School Mathematics for Class 11 & 12*. Bharati Bhawan Publishers. (*Standard higher-secondary pedagogical reference for foundational board preparation and introductory Vieta problems*).
8. **Bernard, S., & Child, J. M.** (1936). *Higher Algebra*. Macmillan and Co. (*Detailed coverage of cubic and biquadratic equations, progression of roots, and reciprocal transformations*).
9. **Lovász, L., Pelikán, J., & Vesztergombi, K.** (2003). *Discrete Mathematics: Elementary and Beyond*. Springer. (*Examines generating functions, power-sum recurrences, and modular arithmetic properties of polynomial roots*).
10. **Barbeau, E. J.** (2003). *Polynomials*. Problem Books in Mathematics. Springer. (*An interactive, problem-driven book containing extensive categories on fractional power transformations, inequalities, and functional equations involving roots*).